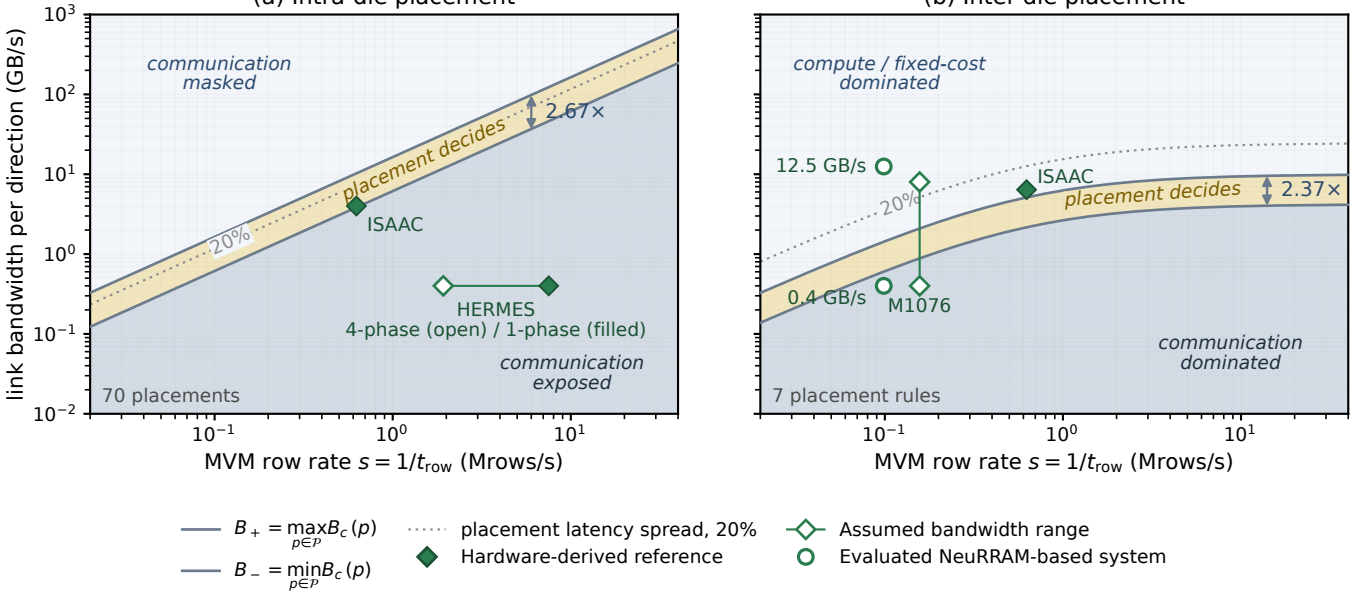
(a) Intra-die placement
(b) Inter-die placement
link bandwidth per direction (GB/s)
MVM row rate $s = 1/t_{\text{row}}$ (Mrows/s)
communication masked
placement decides
2.67×
20%
ISAAC
HERMES
4-phase (open) / 1-phase (filled)
communication exposed
70 placements
compute / fixed-cost dominated
12.5 GB/s
0.4 GB/s
M1076
2.37×
communication dominated
7 placement rules
$B_+ = \max_{p \in \mathcal{P}} B_c(p)$
$B_- = \min_{p \in \mathcal{P}} B_c(p)$
placement latency spread, 20%
Hardware-derived reference
Assumed bandwidth range
Evaluated NeuRRAM-based system